



Health self-management experiences of colorectal cancer patients in postoperative recovery: A qualitative study

Pei-Shan Lo^a, Yu-Ping Lin^b, His-Hsien Hsu^c, Shih-Chang Chang^d, Shu-Ping Yang^e, Wen-Chien Huang^f, Tsae-Jyy Wang^{g,*}

^a Cultivation and Management Center, MacKay Memorial Hospital, New Taipei City, Taiwan, ROC

^b Nursing Department, Oriental Institution of Technology, New Taipei City, Taiwan

^c Department of Colorectal Surgery, MacKay Memorial Hospital, Taipei, Taiwan

^d Division of Colorectal Surgery, Cathay General Hospital, Taipei, Taiwan

^e Nursing Department, Cathay General Hospital, Taipei, Taiwan

^f Department of Thoracic Surgery, Mackay Memorial Hospital, Taiwan

^g School of Nursing, National Taipei University of Nursing and Health Sciences, Taipei, Taiwan, ROC

ARTICLE INFO

Keywords:

Colorectal cancer
Experience
Postoperative
Qualitative study
Self-management

ABSTRACT

Purpose: Health self-management helps improve health-related quality of life and life satisfaction, as well as cancer survival. The study aim was to explore the essence of the health self-management experiences and support needs of colorectal cancer patients after surgeries.

Methods: The study was based on phenomenology research methodology. Purposive sampling was used to obtain a heterogeneous sample to provide rich information regarding the research questions. Participants were recruited from colorectal surgery outpatient clinics in a hospital in Taiwan. Data were collected by semi-structured in-depth interviews and analyzed by thematic content analysis. Strategies adapted from Lincoln and Guba were used to enhance the trustworthiness of the study.

Results: Ten participants, 5 males and 5 females, were interviewed. Their health self-management experience fell into 3 overarching themes and 9 related subthemes. Our results show that (1) seeking support when experiencing discomfort, (2) when life changes, re-adjust accordingly, and (3) staying positive and self-perseverance are the essences of the health self-management experiences and support needs of postoperative colorectal cancer patients.

Conclusions: Postoperative colorectal cancer patients experienced tremendous physical and psychosocial challenges after returning home from the hospital. Although burdened with multiple stressors, these patients were able to seek support, learning to practice self-care, facing cancer positively, and exhibit positive growth in life. Patients with colorectal cancer have to constantly adjust to the impacts of their diseases. The study results may provide as a reference for supporting postoperative adjustment and promoting health self-management among patients with colorectal cancer.

1. Introduction

Colorectal cancer was the third most common cancer worldwide with approximately 1.8 million new cases in 2018 (Bray et al., 2018). Surgery is the primary treatment for colorectal cancer (Dekker et al., 2019; National Comprehensive Cancer Network, 2020). In Taiwan, 4520 adults were diagnosed with colorectal cancer in 2017. Most of them (91.17%) received surgeries as their initial treatments (Taiwan Health

Promotion Administration, 2019). Postoperative patients often face challenges brought on by their disease and treatments. These may include managing physical discomfort and treatment adverse effects, rebuilding daily routines and social roles, preventing disease recurrence, and addressing subsequent psychological distress (Dekker et al., 2019; Dunn et al., 2013a, 2013b; McDermott et al., 2015; Scheele et al., 2015).

Health self-management can be defined as the skills that individuals must learn and practice in the face of chronic health conditions to

* Corresponding author.

E-mail addresses: sami0901kimo@gmail.com (P.-S. Lo), fk014@mail.oit.edu.tw (Y.-P. Lin), hsu5936@ms3.hinet.net (H.-H. Hsu), samson1319@gmail.com (S.-C. Chang), ysppig@yahoo.com.tw (S.-P. Yang), wjhuang0@yahoo.com.tw (W.-C. Huang), tsaejyy@ntunhs.edu.tw (T.-J. Wang).

<https://doi.org/10.1016/j.ejon.2021.101906>

Received 5 April 2020; Received in revised form 28 January 2021; Accepted 2 February 2021

Available online 12 February 2021

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achieve a positive and emotionally satisfying life (Lorig, 1993). Its scope includes health care management, daily life management, and emotional stress management. Health self-management emphasizes partnerships with health care providers to solve problems and perform self-care (Lorig and Holmon, 2003). Good health self-management helps improve health-related quality of life, life satisfaction, and cancer survival (Howell et al., 2016). Understanding the health self-management experiences of patients who have completed surgery and returned home will help develop effective support measures to assist patients in establishing self-management skills for transitioning from the acute stage to long-term survival.

To date, there have been few studies on the self-management experience and needs of postoperative patients with colorectal cancer. Pan et al. (2011) conducted a descriptive, longitudinal study ($n = 35$) to investigate symptom distress and self-care strategies of postoperative colorectal cancer patients with diarrhea. They found that patients rated dietary modification and exercise as the most helpful self-care strategies. In a cross-sectional survey ($n = 175$), Hou et al. (2017) also found that diet self-management was the most common behavior in rectal cancer patients after sphincter-preserving surgery. Most previous studies in understating postoperative colorectal cancer patients' experience have been focused on ostomy associated experience or care needs (Cengiz and Bahar, 2017; Herlufsen and Brodsgaard, 2017; Ceylan and Vural, 2017), the life experience (Jakobsson et al., 2017; Jonsson et al., 2011; Worster and Holmes, 2009), symptoms or distress (Abelson et al., 2018; Krogs-gaard et al., 2014), or the significance of information (Lithner et al., 2015; Samuelsson et al., 2018). Results of an integrative review showed that patients with colorectal cancer faced various experiences during postoperative recovery (Changyai et al., 2020). Suffering, decreased activity, ambivalent feelings, and the need for caring were the main themes of their experiences (Changyai et al., 2020). Only one qualitative study investigated the self-management strategies used by colorectal cancer patients to handle the complicated circumstances within the care of their diseases (Ose et al., 2017). The results showed that "proactive demanding," "proactive behavior," and "proactive data management" were self-management strategies used by patients to handle the complexity of their situations (Ose et al., 2017). However, these authors did specify whether the interviewed patients were in postoperative recovery or other stages of cancer treatment trajectories. Besides, the study findings were generated from focus groups of colorectal cancer patients, representatives of support groups, physicians, and other health care professionals. The results might be different from solely patients' perspectives or inputs.

Therefore, this study uses a phenomenological approach to explore the essence of the health self-management experiences and support needs of colorectal cancer patients during the transition to home after colorectal cancer surgery. Phenomenology research focuses on individuals' experiences of phenomena and how they interpret these experiences (Douglass and Moustakas 1985). For the study purpose, a phenomenological methodology is considered a suitable approach.

2. Methods

2.1. Study design

This is a phenomenology research. Data were collected by semi-structured in-depth interviews and analyzed by thematic content analysis.

2.2. Participants recruitment

Purposive sampling was used to obtain a heterogeneous sample to explore how health self-management is experienced among different patients and provide rich information regarding the research questions. Participants were recruited from colorectal surgery outpatient clinics in a hospital in Taiwan. Potentially eligible patients were referred by

surgeons and approached by one of the researchers (P.-S. L.). Patients who met the following eligibility criteria and could most usefully inform the study and maximize the diversity relevant to the research questions were invited to participate. Inclusion criteria were as follows: (1) aged 20 and older, (2) a confirmed diagnosis of stage I–III colorectal cancer, (3) received tumor resection surgeries in the previous month, and (4) able to speak Mandarin or Taiwanese. Exclusion criteria were as follows: (1) diagnosed with a terminal illness, (2) having a severe mental illness, (3) receiving chemotherapy or radiation therapy pre-operatively, and (4) having a colostomy. The theoretical saturation was used as the criterion for stopping recruitment (Glaser and Strauss, 1967).

2.3. Ethical considerations

The study was approved by the Mackay Memorial Hospital Institutional Review Board (IRB No. 17MMHIS062). One of the researchers (P.-S. L.) explained the study purpose, the interview process, and the tape-recording to the participants. The researcher also assured participants that their data would only be used for research purposes and remained confidential. They could stop the interview or withdraw from the study at any time without penalty or affecting their care. If the interview had caused emotional stress to them, the interviewer would have referred them to the outpatient clinic for assessment and treatment. All participants signed informed consent before the interviews took place. There was neither withdrawal from the study nor adverse reactions to the interviewing occurred.

2.4. Data collection

Data were collected through in-depth interviews by a nursing researcher (P.-S. L.) trained in qualitative methodology. After obtaining a participant's informed consent, the nursing researcher interviewed the participant in a quiet and undisturbed room in the clinic or the participant's home. Data were generated through semi-structured interviews using a topic guide. Interview questions prompted participants to share their experiences, and the interview process also allowed them to express their views and tell their stories. Could you please talk about your experience after returning home from the hospital? How does colorectal cancer affect your life in general? How do you cope with your conditions? What kinds of help have you received or would like to receive from your health care providers? Each interview took about 1–1.5 h and was tape-recorded. The interviewer recorded the interview field notes and retrieved the participants' disease and treatment information from their medical records. All participants were one-to-one interviewed except for one who was interviewed with his wife present.

2.5. Data analysis

Data collection and analysis were carried out simultaneously. Data were analyzed using thematic content analyses. The essence of patients' self-management experience and support needs was explored by understanding and interpreting interview data. The content analysis method is based on existing theoretical knowledge and interprets the data's potential meaning. It can use the categories or themes suggested by a theory to analyze the data in a deductive way. It is also possible to classify data-derived categories or topics when knowledge related to the phenomenon is limited (Hsieh and Shannon, 2005). This study utilized the latter.

After each interview, the recordings were transcribed verbatim and then analyzed. First, two nursing researchers (P.-S. L. and Y.-P. L.) carefully read the verbatim manuscript to understand the interviewee's experience and the overall context and extracted the meaning units for preliminary coding. They discussed and finalized the code list. Next, the two researchers coded the remaining transcripts and organized codes into categories. Third, they organized categories into sub-themes and then organized sub-themes into overarching themes (Table 2). During

Table 1

Demographic and disease characteristics of the study participants.

Case	Sex	Age	Marital	Education	Tumor Sites	Stage	Surgery
1	M	58	Married	Middle	Descending colon	IIIb	Laparoscopic
2	M	52	Single	High	Rectum	IIa	Laparotomy
3	F	44	Divorce	College	Sigmoid colon	I	Laparoscopic
4	M	76	Married	Elementary	Ascending colon	IIIb	Laparotomy
5	F	75	Married	Elementary	Rectum; sigmoid colon	I	Laparoscopic
6	M	51	Married	College	Sigmoid colon	IIIb	Laparotomy
7	M	61	Divorce	Middle	Rectum; sigmoid colon	I	Laparotomy
8	F	59	Married	Middle	Sigmoid colon	IIa	Laparoscopic
9	F	44	Married	College	Sigmoid colon	IIIb	Laparotomy
10	F	59	Married	College	Cecum	I	Laparoscopic

Table 2

The three overarching themes and related subthemes of health self-management experience in patients with colon cancer.

Theme	Subtheme	Definition
Seeking support when experiencing discomfort	1. Attempting to control physical discomfort	The process of attempting to control and improve symptoms when experiencing physical discomfort after the surgical treatment
	2. Trusting physician and cooperating with the treatment plans	Trusting the physician's judgment and cooperating with the physician's directives as best as possible
	3. Searching for information and taking care of oneself	Actively facing the disease and searching for information on treatment, side effects, follow-up, diet, and self-care.
When life changes, re-adjust accordingly	4. Cancer changing one's life and reminding one to pay attention to one's own health	Having experienced the threat of cancer poses to one's life, one realizes the importance of maintaining health.
	5. Reflecting on oneself and changing one's lifestyles	Reexamining how one's own lifestyle has been less than ideal and trying to establish a healthier lifestyle.
	6. Taking care of oneself and cherishing one's family	Learning to care for oneself and not creating a burden for one's family, and recognizing and cherishing the care received from one's family.
Staying positive and self-perseverance	7. Adjusting one's mindset and facing the situation bravely	Looking for coping strategies to face psychological and emotional stress of living with cancer.
	8. Building strength and confidence	Increasing the confidence that one has against cancer by seeking spiritual support and the support of friends and family.
	9. Comprehending life, to learn, and to grow	With the threat of cancer, one realizes the value of life and learns and grows from that experience.

this process, three nursing researchers (P.-S. L., Y.-P. L., and T.-J. W) discussed the data multiple times, and the coding was adjusted if necessary. The results were presented to the participants for verification and feedback.

2.6. Rigor of the study

Strategies adapted from Lincoln and Guba (1986) and Forero et al.

(2018) were used to enhance the trustworthiness of the study in four-dimension criteria (credibility, dependability, confirmability, and transferability). To achieve credibility, we 1) built the research team with all investigators furnished adequate knowledge and skills to accomplish their roles, 2) ensured the interviewer spending time to engage with participants before their interviews, 3) tested the interview protocol with two pilot interviews to finalize the interviewing process and techniques, 4) kept the interview field notes and analyzed them with the interview data, and 5) implemented monthly debriefing sessions with physicians and nurse practitioners of the colorectal surgical department. To achieve dependability, we 1) developed data collection guidelines, 2) kept a detailed record of the data collection process, 3) checked coding accuracy by a second coder to ensure high levels of consistency, and 4) retained the verbatim transcripts and the notes taken during the interviews to allow future verification. To achieve confirmability, the interviewer wrote reflexive journals and discussed it with other researchers to exclude prejudice and bias from an objective and neutral perspective during biweekly team meetings. To achieve transferability, we used a purposive sampling technique to form a designated sample and operationalized data saturation criterion for ceasing recruitment.

3. Results

3.1. Characteristics of participants

Ten participants, five males, and five females were interviewed. Their ages ranged between 44 and 76. The tumor sites included the descending colon, ascending colon, sigmoid colon, cecum, and rectum. The pathological cancer stages ranged from stage I to IIIb (Table 1). Their health self-management experience fell into three overarching themes and nine related subthemes (Table 2).

3.2. Health self-management experience

3.2.1. Theme one: seeking support when experiencing discomfort

This refers to the process of trying new methods, keeping symptoms under control, coping with the treatment, and seeking resources for self-care when experiencing physical discomfort. It includes three sub-themes: "attempting to control physical discomfort", "trusting physician and cooperating with the treatment plans", and "searching for information and taking care of oneself".

"Attempting to control physical discomfort" refers to the process of attempting to control and improve symptoms when experiencing physical discomfort after the surgical treatment. Early in their return, patients often experienced symptoms such as diarrhea, abdominal distension, frequent bowel movements, and loose stool. They tried to develop self-care strategies to improve these symptoms, such as eating smaller meals, walking, having a low fiber diet, and taking prescribed medications. Most of their symptoms gradually improved over time.

"... After I returned home, I still felt some painful flatulence and took medication for it. ... Moving around helps my wounds and bowel

movements recover" (Case 2). *"I experience flatulence, so I don't eat as much"* (Case 7). *"I re-cook food so that it is softer. I have to strain out the fibers in fruits"* (Case 7). *"My husband cuts the vegetables into fine shreds ... I chew slowly and more thoroughly ..."* (Case 8). *"As long as I eat, I have loose stool. However, the number of bowel movements has decreased gradually ..."* (Case 9). *"Needing to use the washroom up to ten times a day has made me wary of leaving the house. He (the physician) has prescribed medication for it, and it seems to help a little"* (Case 5).

"Trusting physician and cooperating with the treatment plans" refers to trusting the physician's judgment and cooperating with the physician's directives as best as possible. Participants indicated that they had limited knowledge of cancer and its treatments, but believed in the physician's expertise, followed the doctor's instructions, and actively cooperated with the treatment plans.

"After the diagnosis, I understood the seriousness of the disease. I cooperated with the doctor – not hiding anything from him. This way, the doctor knew how to treat me" (Case 7). *"The doctor gave a very clear explanation before my surgery. He outlined a couple of likely outcomes and helped me understand the process of the surgery. This explanation made me less anxious and more cooperative with the treatments. Thanks to the care of the doctor and the nurses, I received the best care and arrangements"* (Case 1). *"My doctor is quite clever; I let him arrange things and put my life in his hands"* (Case 5). *"I don't understand the treatment ... I followed the doctor's directives ... my disease healed gradually"* (Case 2).

"Searching for information and taking care of oneself" refers to actively facing the disease and searching for information on treatment, side effects, follow-up, diet, and self-care. Participants wanted to know how to choose and track their diet and searched for relevant information via the hospital's nutrition department, the Internet, and books. Some participants who underwent chemotherapy were concerned about its adverse effects and wished to understand how to bolster their health via their diet or supplements.

"I don't know what foods would be beneficial and what I should be watching out for ... I wish I were provided with this information" (Case 6). *"I now pay more attention to my diet ... I think I got this information from the hospital, and then looked up more information on my phone"* (Case 5). *"My doctor said to eat multiple smaller meals ... I was not sure how small the meals should be, how many meals it should be, but I cut down on the amount ... I am still wondering about the amounts – I don't wish to lose more weight, so I'm worried about eating too little"* (Case 10).

"I don't know to what degree the chemotherapy will affect me, so I'm focused on getting enough nutrition to keep up my strength" (Case 1). *"I just want to rehabilitate my physical strength, and I will feel more at ease while receiving chemotherapy because I heard that chemotherapy seems to be quite uncomfortable"* (Case 6). *"My biggest worry is about recurrence – but I don't know if it will happen, and neither does the doctor ... The doctor said to do a follow-up check every three months but I am not very clear on how or what this would happen"* (Case 10).

3.2.2. Theme two: when life changes, re-adjust accordingly

A cancer diagnosis is a huge shock as well as a turning point in life. Patients realize the importance of their health. They review their past lifestyles and try to establish a healthier diet and exercise routines. They also learn about self-care so that their health can be soon recovered and not cause too much burden to their family. It includes three sub-themes: "cancer changes one's life and reminding one to pay attention to one's health", "reflecting on oneself and changing one's lifestyle", and "taking care of oneself and cherishing one's family".

"Cancer changing one's life and reminding one to pay attention to one's health" refers to having experienced the threat of cancer poses

to life, one realizes the importance of maintaining health. Participants expressed how their cancer diagnosis made them realize the importance of their health and how it caused them to reexamine their previous lack of attention to their health, giving them motives to maintain their health.

"Initially, it was a huge shock. I wondered why I was this unlucky ... This was a turning point in my life, and I reexamined my health ... This is what has happened; you should now learn to cherish yourself and your health" (Case 1). *"I felt that my situation was caused by not taking care of my health"* (Case 2). *"I am still quite young; I don't want to be bedbound ... Now I am focused on regaining my health"* (Case 7). *"One must maintain one's health and not rely too much on others ... We cannot schedule illnesses and prepare for them in advance"* (Case 10).

"Reflecting on oneself and changing one's lifestyles" refers to reexamining how one's lifestyle has been less than ideal and trying to establish a healthier lifestyle. Most participants reported reviewing their unhealthy diets and lifestyle habits and noted that they required self-discipline and a healthier lifestyle. This would involve diet, exercise, smoking, alcohol, and stress from work.

"After I was diagnosed with colon cancer, I felt that I should have led a more balanced lifestyle so that this would not have happened" (Case 2). *"Because of my work, I often ate out. This might have contributed to my tumor"* (Case 6). *"After the surgery, I had to cooperate with the treatment plan and change my old habits. If I did not change them, my body would be like it had been ... Now I don't drink or smoke anymore and eat three regular meals"* (Case 2). *"When I hear that some food will help, or some doctor will help, I pursue these options to my utmost"* (Case 10). *"One must think of ways to change bad habits ... I need to remember to eat healthier, exercise regularly, and find ways to relieve pressure ..."* (Case 3). *"I need to force myself to relax. Sometimes I get too anxious, sleep too late, and that needs to be changed ..."* (Case 10). *"Living is more important; it's alright if work performance suffers. Before, I would ask more of myself and put a lot of pressure on myself, hoping to make everything perfect"* (Case 3). *From now on, I know that I need to get regular checkups because prevention is far better than treatment"* (Case 1).

"Taking care of oneself and cherishing one's family" refers to learning to care for oneself and not creating a burden for one's family, and recognizing and cherishing the care received from one's family. Most participants tried to take care of themselves to prevent burdening their families. Their family and friends' love and support made these patients more motivated to take care of themselves. They tried not to make their family worry and not to be a burden on them.

(Crying) "I feel like a waste. I am a burden to my family. My mother, who is already in her seventies, has to take care of me" (Case 9). *"I don't wish to affect my family, especially my mother, because I am her only son"* (Case 6). *"I bear it all myself; I don't wish to disrupt her (the daughter's) life"* (Case 5). *"My wife did all she could to take care of me. I'm very grateful to her"* (Case 1). *"I listened to my doctor and did whatever they said ... The faster my body heals, the less I worry my family ..."* (Case 5). *No matter the disease, it will cause some burden to the household ... We need to cherish our health – loving ourselves is cherishing our family ... When one is healthy, it is easier for the entire family"* (Case 1).

3.2.3. Theme three: staying positive and self-perseverance

Diagnosed with cancer causes emotional turbulence, which one should face with coping mechanisms and a positive attitude. Spiritual and social support relieve the negativity of one who is feeling helpless. One learns and grows from the experience, allowing one to persevere. The three subthemes are: "adjusting one's mindset and facing the situation bravely", "building strength and confidence", and "comprehending life, to learn, and to grow from the experience".

“Adjusting one’s mindset and facing the situation bravely” refers to looking for coping strategies to face the psychological stress and emotional turbulence of living with cancer. Participants reported searching for ways to cope with the stress and adjust their attitude, including searching for communication and ways to vent, maintaining optimism, keeping a normal outlook, and letting go of the pressure.

“I felt depressed, so my sister told me to write down my negative emotions ... When I shift my focus, those feelings pass quickly” (Case 9). *“It’s like an untimed bomb ... Initially, I couldn’t shake the possibility of a recurrence, so I felt stressed ... but I must adjust my attitude”* (Case 3). *“I let nature take its course and don’t overthink about it. I am old and don’t worry too much about it – if it is my time, it is my time; if it is not, then I stay”* (Case 4). *“It happened – worrying, crying, and being sad doesn’t help at all; one must face it bravely ... listen to some poems and music ... make peace with it, or try not to be alone because being alone with nothing to do makes my imagination run wild ...”* (Case 1). *“I saw some patients who have a more severe case and require an artificial stoma ... I feel that I am already luckier than they are”* (Case 6). *“I will get better eventually ... I am very optimistic and always think towards the pros ...”* (Case 5). *“One must adjust ... If I did not set aside those feelings, my family would notice my mood and become worried ... I walked around, and my spirits lifted. Constantly sitting does no good to one’s spirit”* (Case 8). *“I’ve done all that I could ... I focused on getting better ... that is an active treatment”* (Case 10).

“Building strength and confidence” refers to increasing the confidence that one has against cancer by seeking spiritual support and the support of friends and family. Participants reported feeling bolstered by their faith and by the people around them to face cancer bravely.

“I believe in God – I put everything in His hands; it is His plan” (Case 6). *“I gave all my worries to God ... He will protect me ... I have felt very peaceful, and am full of God’s grace I have God to rely upon. His plans are superior to my plans; His wisdom is superior to mine. I do my part, which is to have regular checkups and follow-ups, to undergo chemotherapy”* (Case 1). *“I used to pray to idols; now I have seen the light and have turned to God ... Some of my sisters in Christ contact me often and give me warmth, and make me feel better”* (Case 5). *“My daughter was supportive. She told me that it was alright; they removed the entire tumor ... Initially, I was panicking, but I encouraged myself to stay strong, that it would all be alright”* (Case 5). *“It is the first time that I have been ill. My husband is very caring, and so are my children”* (Case 10). *“I believe that one feels better with the support of family and friends because one feels that someone is helping”* (Case 3).

“Comprehending life, to learn, and to grow.” With the threat of cancer, one realizes the value of life and learns and grows from that experience. Participants reported that with the danger of cancer and their life changes, they reexamined the pricelessness of life and replanned their goals for the future. They achieved spiritual growth, having been reborn from the ashes, and now have a more determined life outlook.

“Having been through this, I know how to bring others warmth ... I want to help people” (Case 6). *“My sisters in Christ gave me a lot of warmth ... I hope that after I am better, I will go out and spread His word with my sisters”* (Case 5). *“Follow the footsteps of the Lord and do His work, comforting people ...”* (Case 1).

4. Discussion

Our results show that (1) seeking support when experiencing discomfort, (2) when life changes, re-adjust accordingly, and (3) staying positive and self-perseverance are the essences of the health self-management experiences and support needs of postoperative colorectal cancer patients.

4.1. Seeking support when experiencing discomfort

Similar to previous studies’ findings (Dekker et al., 2019; Hildebrandt et al., 2019), diarrhea and bloating were common symptoms experienced in these postoperative colorectal cancer patients. In attempting to control physical discomfort, these participants tried to eat small meals and make dietary adjustments, as reported in previous studies. However, the physical symptoms can still adversely affect an individual’s daily life (Russell et al., 2015). Some participants reported troubled by frequent bowel movements and were reluctant to go out because of the inconvenience and the embarrassment of frequently needing a toilet. As for the subtheme, trusting physicians and cooperating with the treatment, participants indicated that they rely on physicians’ professional judgment while facing cancer diagnosis challenges and the uncertainty of its treatment outcomes. Similar to what was reported in a previous study (Lam et al., 2014), patients felt better prepared psychologically. They were more able to actively cooperate with their treatment plans after discussing with their physicians and receiving medical professionals’ explanations. They try to follow their physicians’ suggestions and cooperate fully with the treatment plans in the hopes of recovering. Participants realized the importance of health after been diagnosed with cancer and searched for self-care information to manage their conditions. Some participants felt that the health information provided by their medical teams was not sufficient and wished for more precise information on dietary and follow-up suggestions. As suggested in previous studies, patients reported great needs for self-care information after cancer treatment and hoped to receive relevant information from their medical teams (Hoekstra et al., 2014).

4.2. When life changes, re-adjust accordingly

Participants perceived the threat of cancer to their life and realized the importance of maintaining health as the subtheme, cancer changes one’s life and causes one to face one’s health. Participants reexamined their health practice and tried to maintain their health better. Similar to what was reported in previous studies (Hildebrandt et al., 2019; McLachlan et al., 2015), a cancer diagnosis was seen as a warning for an individual to pay more attention to their health. Relevant education and support measures can be used to enhancing patients’ self-care during this critical period. As for the subtheme, reflecting on oneself and changing one’s lifestyle, participants thought of their previous unhealthy lifestyles as the cause of their cancer. They tried to maintain a healthy lifestyle to decrease the risk of recurrence. Participants reported searching for information about food choices, physical activity, and dietary supplements. Similar to findings in previous studies, cancer patients often linked the cause of cancer with their unhealthy behavior and decided to work hard at changing their lifestyles (Hildebrandt et al., 2019; Kanera et al., 2016; McLachlan et al., 2015). Rock et al. (2012) also reported that cancer survivors often seek self-care strategies to improve their long-term outcomes, reduce recurrence risk, and improve quality of life. This could be treated as a window of opportunity to help patients build a healthy lifestyle. As for the subtheme, taking care of oneself and cherishing one’s family, participants experienced the care that their family provided during their treatment and cherished their feelings and connections with their families. Participants expressed a strong desire to recover quickly through self-care as a way to repay their family. Similar to what was reported in a qualitative study (Hildebrandt et al., 2019; Law et al., 2018), perceived support from family and important others affirmed the value of life and enhanced patients’ cooperation with the treatment plans. This gratefulness is also a motivator for adopting a healthy lifestyle.

4.3. Stay positive and genuine to yourself

After surgical resection, patients were troubled by symptoms and under great emotional and psychological stress. They must face their

situations with a positive attitude and coping strategies. Therefore, the participants reported the experience of adjusting one's perspective and facing the crisis bravely. As findings in previous studies, the study participants said that positivity and optimistic attitudes help them meet the psychological challenges of colorectal cancer (Dunn et al., 2013b; Seiler and Jenewein, 2019). These findings suggest the importance of resilience for psychological adjustment in this population because resilience is the ability to show positive emotions, such as hope, optimism, and happiness, and change the crisis into a turning point (Seiler and Jenewein, 2019). As for the subtheme, building strength and confidence, the participants were able to face cancer courageously with the help of religious beliefs, support from family and friends, and the sharing of experience among patients. Similar to previous studies' findings, religious belief and support systems helped patients face the psychosocial challenges of cancer (Chang et al., 2016; Law et al., 2018; Saraf et al., 2013; Whitford and Olver, 2012). The study findings also support that family is a core support group and plays a vital role in supporting the psychological adjustment of cancer patients (Law et al., 2018). As for the subtheme, comprehending life, to learn and to grow, the life-threatening experience of having cancer remind a patient to think about the value of life. Some participants took it as an opportunity to re-plan their life and restart again. Some were more involved in religion. Others appreciated their family and friends' support and hoped that they could help those in need afterward. Similar to previous studies' findings (Saraf et al., 2013; Whitford and Olver, 2012), cancer patients might redefine their meaning of life and become more active in their beliefs. These may enhance their spiritual growth and inspire them to care for others, thereby improving their life value.

4.4. Limitations and recommendations for future research

There are several limitations to this study. First, we recruited only patients who received tumor resection surgeries in the previous month. These patients can only provide acute postoperative self-management experience, not the complete picture of self-management experience. In particular, all patients in this stage had not yet received any adjuvant chemotherapy or radiotherapy, which was usually started later on. Patients might require different self-management strategies during chemotherapy or radiotherapy. Second, we excluded patients receiving chemotherapy or radiation therapy pre-operatively, or with a colostomy; therefore, these patients' experience was not included in the current study. Third, interviews took place when participants returned to the hospital for a follow-up appointment and mostly occurred in the hospital's prayer room. The timing and locations might not be ideal for interviewing and may impede participants from expressing themselves fully. Moreover, the presence of a recording or a family member may also hinder the depth of interviews. Two participants expressed their concerns regarding the tape recording and would only speak more freely after the interviewer promised to maintain their confidentiality. One participant was interviewed with his wife present. Nevertheless, the study results help us to understand the health self-management experience of colorectal cancer surgical patients.

Future studies exploring patients' self-management experience receiving chemotherapy or radiotherapy, or those with a colostomy, can advance our knowledge in these patients. They might have different concerns and need a different set of self-management skills and support. A longer duration follow-up is suggested for understanding changes in patients' self-management experience overtime. One may also consider using the grounded theory approach to explore the social interaction and process of developing self-management competency after colorectal cancer surgery.

4.5. Relevance to clinical practice

The study findings are relevant to understanding patients' health self-management experience during the transition to home after

colorectal cancer surgery. The patients indicated a strong desire for information on dietary choices, follow-up methods, and physical strength maintenance during chemotherapy. However, most patients reported that nurses' care information and the content of the health education manual were not specific enough. These findings suggest areas for nurses to improve and advance their practice. Although the period in the hospital may be very brief for these patients, nurses can provide tailored discharge instructions that address self-management of physical symptoms and emotional distress. Nurses or nurse practitioners in the outpatient area, who at the first follow-up visit, should assess patients for issues and intervene appropriately. Furthermore, patients reported that the support of medical personnel, religion, family and friends, and fellow patients helped them stay positive and be confident that they could overcome cancer. Cancer care is multidimensional and holistic. Nurses should develop a set of plans by which patients receive complete medical care and support and assistance from professionals and family members as their treatment progresses to help patients face cancer challenges. The study results provide a reference for nurses to support postoperative adjustment and promote health self-management among patients with colorectal cancer.

5. Conclusion

The immediate transition to home after colorectal cancer surgery presents the patient and family with multiple physical and psychological challenges and requires a shift in self-management strategies. This phenomenological research explored the essence of patients' health self-management experience during this critical transition. The results showed that patients would "seek support when experiencing discomfort," searching for resources and cooperating with their treatment. They also felt the threat that cancer posed to their lives and its effect on their health. They realized that "when life changes, re-adjust accordingly," and reviewed their past behaviors as well as appreciated the care provided by their family. Despite the shock of their diagnosis, they felt supported and tried to "stay positive and true to themselves," searching for coping mechanisms and appreciating the value of life. Patients with colorectal cancer have to adjust to the impacts of their diseases and treatments constantly. Although burdened with multiple stressors, patients could seek support, learn to practice self-care, face cancer positively, and exhibit positive growth in life.

CRediT authorship contribution statement

Pei-Shan Lo: study design, data collection, data analysis, manuscript preparation. **Yu-Ping Lin:** study design, data analysis, and manuscript preparation. **His-Hsien Hsu:** study design, manuscript preparation. **Shih-Chang Chang:** study design, manuscript preparation. **Shu-Ping Yang:** data collection, data analysis. **Wen-Chien Huang:** study design, manuscript preparation. **Tsae-Jyy Wang:** study design, data analysis, and manuscript preparation.

Declaration of competing interest

None declared.

Acknowledgements

The authors thank all the participants in this study. The study is supported by a funding of the Ministry of Science and Technology, Taiwan (MOST 106-2314-B-227-003 -).

References

- Abelson, J.S., Chait, A., Shen, M.J., Charlson, M., Dickerman, A., Yeo, H.L., 2018. Sources of distress among patients undergoing surgery for colorectal cancer: a qualitative study. *J. Surg. Res.* 226, 140–149. <https://doi.org/10.1016/j.jss.2018.01.017>.

- Bray, F., Ferlay, J., Soerjomataram, I., Siegel, R.L., Torre, L.A., Jemal, A., 2018. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA A Cancer J. Clin.* 68, 394–424. <https://doi.org/10.3322/caac.21492>.
- Cengiz, B., Bahar, Z., 2017. Perceived barriers and home care needs when adapting to a Fecal Ostomy: a phenomenological study. *J. Wound, Ostomy Cont. Nurs.* 44 (1), 63–68. <https://doi.org/10.1097/WON.0000000000000271>.
- Ceylan, H., Vural, F., 2017. Living with stoma – a phenomenological study. *J. Nurs. Res. Pract.* 1 (1), 6–12. <https://doi.org/10.1016/j.apnr.2017.10.005>.
- Chang, M.Y., Hsu, W.Y., Hsu, C.Y., Lin, L.Y., 2016. Quality of life and related factors in patients with newly diagnosed cancer. *VGH Nurs.* 33, 293–302. <https://doi.org/10.6142/VGHN.33.3.293>.
- Changyai, K., Kongvattananon, P., Somprasert, C., 2020. The experiences of colorectal cancer patients in postoperative recovery: integrative review. *J. Health Res.* 34 (3), 259–269. <https://doi.org/10.1108/JHR-05-2019-0103>.
- Dekker, E., Tanis, P.J., Vleugels, J.L.A., Kasi, P.M., Wallace, M.B., 2019. Seminar colorectal cancer. *Lancet* 394, 1467–1480. [https://doi.org/10.1016/S0140-6736\(19\)32319-0](https://doi.org/10.1016/S0140-6736(19)32319-0).
- Douglas, B.G., Moustakas, C., 1985. Heuristic inquiry: the internal search to know. *J. Humanist. Psychol.* 25 (3), 39–55. <https://doi.org/10.1177/0022167885253004>.
- Dunn, J., Ng, S.K., Breitbart, W., Aitken, J., Youl, P., Baade, P.D., Chambers, S.K., 2013a. Health-related quality of life and life satisfaction in colorectal cancer survivors: trajectories of adjustment. *Health Qual. Life Outcome* 11, 46. <https://doi.org/10.1186/1477-7525-11-46>.
- Dunn, J., Ng, S.K., Holland, J., Aitken, J., Youl, P., Baade, P.D., Chambers, S.K., 2013b. Trajectories of psychological distress after colorectal cancer. *Psycho Oncol.* 22 (8), 1759–1765. <https://doi.org/10.1002/pon.3210>.
- Forero, R., Nahidi, S., Costa, J.D., Mohsin, M., Fitzgerald, G., Gibson, N., McCarthy, S., Aboagye-Sarfo, P., 2018. Application of four-dimension criteria to assess rigour of qualitative research in emergency medicine. *BMC Health Serv. Res.* 18, 120. <https://doi.org/10.1186/s12913-018-2915-2>.
- Glaser, B.G., Strauss, A.L., 1967. *The Discovery of Grounded Theory: Strategies for Qualitative Theory*. Aldine Transaction, New Brunswick.
- Herlufsen, P., Brodsgaard, A., 2017. The lived experiences of persons hospitalized for construction of an urgent Fecal Ostomy. *J. Wound, Ostomy Cont. Nurs.* 44 (6), 557–561. <https://doi.org/10.1097/WON.0000000000000382>.
- Hildebrandt, C., Mayer, H., Koller, A., 2019. Experiences of patients with colorectal cancer from diagnosis until completion of treatment: a meta-ethnography approach. *Psycho Oncol.* 28 (2), 219–227. <https://doi.org/10.1002/pon.4946>.
- Hoekstra, R.A., Heins, M., Korevaar, J.C., 2014. Health care needs of cancer survivors in general practice: a systematic review. *BMC Fam. Pract.* 15 (1), 94. <https://doi.org/10.1186/1471-2296-15-94>.
- Hou, X.T., Pang, D., Lu, Q., Yang, P., Jin, S.L., 2017. Bowel dysfunction and self-management for bowel symptoms after sphincter-preserving surgery: a cross-sectional survey of Chinese rectal cancer patients. *Canc. Nurs.* 40 (4), E9–E16. <https://doi.org/10.1097/NCC.0000000000000393>.
- Howell, D., Harth, T., Brown, J., Bennett, C., Boyko, S., 2016. *Self-management Education for Patients with Cancer: Evidence Summary*. Cancer Care Ontario, Toronto (ON), pp. 20–23.
- Hsieh, H.F., Shannon, S.E., 2005. Three approaches to qualitative content analysis. *Qual. Health Res.* 15 (9), 1277–1288. <https://doi.org/10.1177/1049732305276687>.
- Jakobsson, J., Idvall, E., Kumlien, C., 2017. The lived experience of recovery during the first 6 months after colorectal cancer surgery. *J. Clin. Nurs.* 26 (23–24), 4498–4505. <https://doi.org/10.1111/jocn.13780>.
- Jonsson, C.A., Stenberg, A., Frisman, G.H., 2011. The lived experience of the early postoperative period after colorectal cancer surgery. *Eur. J. Canc. Care* 20 (2), 248–256. <https://doi.org/10.1111/j.1365-2354.2009.01168.x>.
- Kanera, I.M., Bolman, C.A., Mesters, I., Willems, R.A., Beaulen, A.A., Lechner, L., 2016. Prevalence and correlates of healthy lifestyle behaviors among early cancer survivors. *BMC Canc.* 16 (1), 4. <https://doi.org/10.1186/s12885-015-2019-x>.
- Krogsgaard, M., Dreyer, P., Egerod, I., Jarden, M., 2014. Post-discharge symptoms following fast-track colonic cancer surgery: a phenomenological hermeneutic study. *SpringerPlus* 3 (276), 1–8. <https://doi.org/10.1186/2193-1801-3-276>.
- Lam, W.W., Kwok, M., Chan, M., Hung, W.K., Ying, M., Or, A., Kwong, A., Suen, D., Yoon, S., Fielding, R., 2014. Does the use of shared decision-making consultation behaviors increase treatment decision-making satisfaction among Chinese women facing decision for breast cancer surgery? *Patient Educ. Counsel.* 94 (2), 243–249. <https://doi.org/10.1016/j.pec.2013.11.006>.
- Law, E., Levesque, J.V., Lambert, S., Girgis, A., 2018. The "sphere of care": a qualitative study of colorectal cancer patient and caregiver experiences of support within the cancer treatment setting. *PLoS One* 13 (12), e0209436. <https://doi.org/10.1371/journal.pone.0209436>.
- Lincoln, Y.S., Guba, E.G., 1986. But is it rigorous? Trustworthiness and authenticity in naturalistic evaluation. *N. Dir. Progr. Eval.* 30, 73–84. <https://doi.org/10.1002/ev.1427>.
- Lithner, M., Klefsjard, R., Johansson, J., Andersson, E., 2015. The significance of information after discharge for colorectal cancer surgery – a qualitative study. *BMC Nurs.* 14 (36), 1–8. <https://doi.org/10.1186/s12912-015-0086-6>.
- Lorig, K., 1993. Self-management of chronic illness: a model for the future. *Generations* 17 (3), 11–14. <https://www.jstor.org/stable/44877774>. (Accessed 3 April 2020).
- Lorig, K.R., Holman, H.R., 2003. Self-management education: history, definition, outcomes, and mechanisms. *Ann. Behav. Med.* 26 (1), 1–7. https://doi.org/10.1207/S1532796ABM2601_01.
- McDermott, F.D., Heeney, A., Kelly, M.E., Steele, R.J., Carlson, G.L., Winter, D.C., 2015. Systematic review of preoperative, intraoperative and postoperative risk factors for colorectal anastomotic leaks. *Br. J. Surg.* 102 (5), 462–479. <https://doi.org/10.1002/bjs.9697>.
- McLachlan, S., Mansell, G., Sanders, T., Yardley, S., van der Windt, D., Brindle, L., Chew-Graham, C., Little, P., 2015. Symptom perceptions and help-seeking behaviour prior to lung and colorectal cancer diagnoses: a qualitative study. *Fam. Pract.* 32 (5), 568–577. <https://doi.org/10.1093/fampra/cmv048>.
- National Comprehensive Cancer Network, 2020. Clinical Practice Guidelines in Oncology (NCCN Guidelines®), Colon Cancer. Version 2.2020. https://www.nccn.org/practiceguidelines/physician_gls/pdf/colon.pdf. (Accessed 24 March 2020).
- Ose, D., Winkler, E.C., Berger, S., Baudendistel, I., Kamradt, M., Eckrich, F., Szecsenyi, J., 2017. Complexity of care and strategies of self-management in patients with colorectal cancer. *Patient Prefer. Adherence* 11, 731–742. <https://doi.org/10.2147/PPA.S127612>.
- Pan, L.H., Tsai, Y.F., Chen, M.L., Tang, R., Chang, C.J., 2011. Symptom distress and self-care strategies of colorectal cancer patients with diarrhea up to 3 months after surgery. *Canc. Nurs.* 34 (1), E1–E9. <https://doi.org/10.1097/NCC.0b013e3181e3ca21>.
- Rock, C.L., Doyle, C., Demark-Wahnefried, W., Meyerhardt, J., Courneya, K.S., Schwartz, A.L., Bandera, E.V., Hamilton, K.K., Grant, B., McCullough, M., Byers, T., Gansler, T., 2012. Nutrition and physical activity guidelines for cancer survivors. *CA A Cancer J. Clin.* 62 (4), 242–274. <https://doi.org/10.3322/caac.21146>.
- Russell, L., Gough, K., Drosowsky, A., Schofield, P., Aranda, S., Butow, P.N., Westwood, J.A., Krishnasamy, M., Young, J.M., Phipps-Nelson, J., King, D., Jefford, M., 2015. Psychological distress, quality of life, symptoms and unmet needs of colorectal cancer survivors near the end of treatment. *J. Canc. Surv.* 9 (3), 462–470. <https://doi.org/10.1007/s11764-014-0422-y>.
- Samuelsson, K.S., Egenvall, M., Klarin, I., Lokk, J., Gunnarsson, U., Iwarzon, M., 2018. The older patient's experience of the healthcare chain and information when undergoing colorectal cancer surgery according to the enhanced recovery after surgery concept. *J. Clin. Nurs.* 27 (7–8), e1580–e1588. <https://doi.org/10.1111/jocn.14328>.
- Saraf, S., Singh, T.B., Khurana, S., 2013. Cervical cancer survivors: meaning in life. *Psychol. Stud.* 58 (2), 144–152. <https://doi.org/10.1007/s12646-013-0186-x>.
- Scheele, J., Lemke, J., Meier, M., Sander, S., Sander, S., Henne-Bruns, D., Kornmann, M., 2015. Quality of life after sphincter-preserving rectal cancer resection. *Clin. Colorectal Canc.* 14 (4), 33–40.
- Seiler, A., Jenewein, J., 2019. Resilience in cancer patients. *Front. Psychiatr.* 10, 208. <https://doi.org/10.3389/fpsy.2019.00208>.
- Taiwan Health Promotion Administration, 2019. Cancer registry annual report for 2017 in Taiwan. Retrieved from <https://www.hpa.gov.tw/Pages/Detail.aspx?nodeid=269&pid=12235>.
- Whitford, H.S., Olver, I.N., 2012. The multidimensionality of spiritual wellbeing: peace, meaning, and faith and their association with quality of life and coping in oncology. *Psycho Oncol.* 21 (6), 602–610. <https://doi.org/10.1002/pon.1937>.
- Worster, B., Holmes, S., 2009. A phenomenological study of the postoperative experiences of patients undergoing surgery for colorectal cancer. *Eur. J. Oncol. Nurs.* 13 (5), 315–322. <https://doi.org/10.1016/j.ejon.2009.04.008>.